Module 14 — Case Studies: Messaging and Media

Module Introduction

What this module is. Five more complete designs, built the same way as the Module 12 capstones: numbers first, then the five questions, then boxes, deep dives, and a walk through every failure. No new patterns are introduced. Every mechanism used here already has its own lesson in Modules 01–11 or a chapter in the DDIA notes (DDIA = *Designing Data-Intensive Applications*, Martin Kleppmann's book on how data systems work; this repository has chapter-by-chapter notes on it), and this module links to them instead of re-teaching them. What is new is the *shape* of the problems: all five move either **messages between people** or **large media files to many screens**.

> **Analogy.** Module 12 was four complete chess games. This module is five more games that all start from the same opening (a client that must stay in step with a server), but branch into very different middle games: tiny messages that must arrive in order, letters that must never be lost, films that must be cooked once and served a billion times, folders that must look identical on five devices, and a broadcast that has to reach twenty million people within seconds.

The five designs

Case	Center of gravity (the main problem)	What decides the architecture
1. Chat System	Ordered, deduplicated (no repeats) delivery to users whose sockets (open network connections) live on thousands of machines	Connection count and per-conversation ordering
2. Email Service	"Never lose an accepted letter" while talking to an untrusted, slow internet	The SMTP (email protocol) durability boundary and deliverability (getting into inboxes, not spam)
3. Video Platform	Pay once to transcode (convert into other formats and sizes), then serve a power-law audience (a few videos get most views) from the edge (servers near viewers)	Egress bandwidth (tens of Tbps, terabits per second) and encode compute
4. File Sync & Storage	Many devices editing one tree, offline, with no silent data loss	Chunk-level dedupe (store identical pieces once) and the change journal (log of every change)
5. Live Streaming	Seconds of latency (delay) to millions of simultaneous viewers of one stream	The latency budget and edge fan-out of one hot object

Four terms used throughout (each is taught properly in the linked lesson):

- **Fan-out:** one input producing many deliveries (one chat message to 500 group members; one video segment to 20M viewers). See the news feed in Module 12.
- **Cursor:** a saved position in an ordered log ("I have seen everything up to sequence 4,812"). Clients sync by asking for "everything after my cursor". See Cursor-Based Pagination and DDIA's partitioned logs.
- **Blob vs metadata:** the big immutable bytes (a video file, an attachment, a file chunk) versus the small mutable facts about them (title, owner, labels, which chunks make up version 7). Every case in this module stores the two in different systems.
- **Egress:** bytes leaving your servers towards users. For media systems it is the biggest bill and the reason CDNs exist (CDN). A **CDN** (content delivery network) is a worldwide fleet of cache

> servers that keep copies of files close to users, so most bytes never travel from your own data centre.

If words like latency, throughput, cache, replication, sharding or load balancer are new to you, read System Design Fundamentals first; this module assumes them.

How to use this module. Same routine as Module 12: at the start of each case, cover the page below the prompt and do stages 1–3 yourself on paper (numbers, five questions, first boxes). Then read on and compare. The cases are independent; do the one closest to your next interview first. Chat and live streaming share connection-tier ideas; video and live streaming share the CDN and adaptive-bitrate ideas; email and file sync share the blob/metadata split.

The stages used in every case (Module 12's seven stages, with API and data model made explicit because interviewers ask for them in these problems):

1. Requirements and numbers
2. The five-questions pass
3. API
4. Data model
5. The architecture, annotated
6. Deep dives
7. The failure walk (scaling and failure)
8. Trade-offs
9. The follow-up questions
10. Pattern manifest and TL;DR

All scale figures are **assumptions chosen for the exercise**, rounded so the arithmetic is easy. Where a real company has published how it does something, the case says so and cites the company's own engineering write-up.

Case 1 — Chat System

The prompt (in short). Build a messaging app like WhatsApp, Messenger or Slack DMs: one-to-one chats and group chats (up to 1,000 members), messages delivered in real time when the recipient is online and on reconnect when they are not, history on every device, read receipts, and online presence.

What makes it hard. The message itself is tiny (a few hundred bytes). The hard parts are: (1) **hundreds of millions of open connections** spread across thousands of machines, so "send to Bob" is a routing problem; (2) every participant must see a conversation **in the same order**, without gaps or duplicates, even though phones drop off networks constantly; and (3) the same user has **several devices** that must all converge on the same history.

1. Requirements and numbers

Functional: send/receive 1:1 and group messages (text plus media links), delivery while online and catch-up after being offline, full history on all of a user's devices, delivered/read receipts, presence ("online", "last seen"), push notifications when the app is closed.

Non-functional: end-to-end delivery **p99 < 500 ms** when both users are online (**p99** = the 99th percentile: 99 out of 100 messages must be faster than this; ms = milliseconds); **no lost acknowledged messages**; per-conversation **total order** (everyone sees the same order); duplicates are never shown; high availability (the service is up and answering almost all the time; a chat outage is front-page news).

Numbers (assumptions):

- **500M daily active users (DAU)**, each sending **40 messages/day → 20B messages/day ≈ 230K messages/s average, ~700K/s peak** (3× average). (B = billion, K = thousand, M = million; "/s" = per second.)
- Average recipients per message: **~3** (most chats are 1:1, some groups) → **~2M deliveries/s at peak**.
- Message size including metadata ≈ **300 bytes** → 20B × 300 B = **6 TB/day**, **~2.2 PB/year** before replication (×3 → ~6.6 PB/year). (1 TB, terabyte, = 1,000 GB; 1 PB, petabyte, = 1,000 TB. **Replication** = keeping several copies on different machines so a dead disk loses nothing; three copies is common.) History is kept forever, so storage grows linearly and **old data is almost never read**.
- Peak concurrent connections: ~30% of DAU online at once = **150M sockets**. Assume a conservative **50K sockets per gateway** (a **gateway** here is a server whose only job is to hold client connections open) (a tuned server can hold more, but smaller gateways mean fewer users reconnect when one restarts) → **3,000 gateway servers**. The gateway tier itself is the one described in Bidirectional Streaming.
- Heartbeats (tiny "I'm still here" messages that prove a connection is alive; also called keepalives or pings): 150M sockets × one ping every 30 s = **5M pings/s**. These must be answered **at the gateway** and never touch the backend; this one number rules out designs where every keepalive hits a database.

What the numbers say: throughput per message is moderate; the **connection count** and the **write-mostly, append-only history** (new data is only ever added at the end, never changed in place) decide the architecture.

2. The five-questions pass

- **Flow:** the sender's app sends over its WebSocket (a long-lived, two-way connection between app and server, so either side can send at any time) and waits only for a server ack (acknowledgement: "stored, sequence 812"). Delivery to recipients is **asynchronous** (it happens in the background; the sender does not wait for it). Offline recipients get a push notification and fetch on reconnect (Five Questions).
- **Storage:** the **message log per conversation** is the source of truth (the one copy that wins if others disagree). Everything else (unread counts, conversation list, search index) is derived (computed from it) and can be rebuilt (CQRS).
- **Speed:** the hottest read is "the last 50 messages of this conversation" on app open; it must be one partition read (a **partition** is one slice of the data, stored together on one machine, so reading it needs one machine).
- **Failure:** networks drop constantly on mobile. Design for **at-least-once delivery + dedupe** (send again until confirmed, which may produce repeats, then throw repeats away at the receiver), never for a perfect connection (Idempotency).
- **Growth:** connections grow with users; storage grows with time; hot spots are big groups and bursts (New Year's Eve).
- *Skeleton:* stateless-ish connection gateways (they keep no data that matters if they die), a session registry, a per-conversation sequencer, an append-only message store partitioned by conversation, and a per-user sync cursor.

3. API

Real-time traffic uses a small protocol over one WebSocket per device; everything bulky uses plain HTTPS (ordinary encrypted web requests).

Direction	Frame / endpoint	Purpose
Client → server	`send {client_msg_id, conv_id, body, media_ref?}`	`client_msg_id` is a UUID (random 128-bit ID) the client generates once and reuses on every retry (an idempotency key: lets the server spot a repeat)
Server → sender	`ack {client_msg_id, msg_id, seq, server_ts}`	"Stored durably; your message is number `seq` in this conversation"
Server → recipient	`msg {conv_id, seq, msg_id, sender, body, server_ts}`	Live delivery
Client → server	`receipt {conv_id, up_to_seq, kind: delivered/read}`	One receipt covers everything up to `seq` (cumulative, not per message)
HTTPS	`GET /sync?cursor=…`	Catch-up after reconnect: all changes after the device's cursor
HTTPS	`GET /conversations/{id}/messages?before_seq=…&limit=50`	Scroll back through history (cursor pagination)
HTTPS	`POST /media` → pre-signed upload URL (temporary, signed link allowing one upload)	Photos/videos go to blob storage (a store for big files), never through the socket

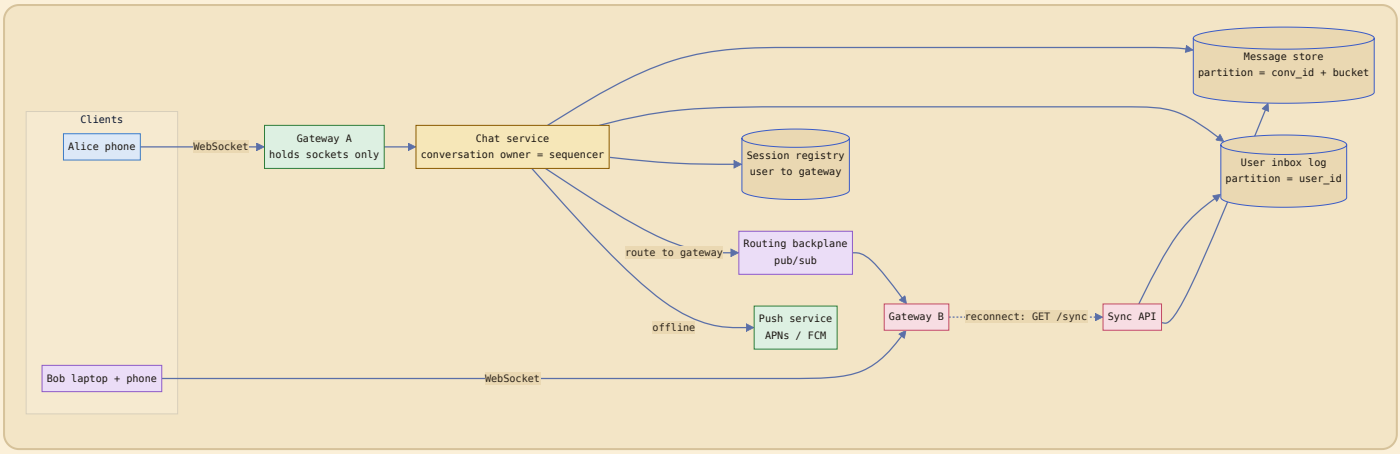
4. Data model

- **`messages`** — partition key `(conv_id, bucket)`, clustering key `seq` descending. A **bucket** is a time or size slice (for example 10 days, or 100K messages) so that one busy conversation never becomes one unbounded partition. Columns: `msg_id, sender_id, body, media_ref, server_ts, flags (edited/deleted)`. An LSM-tree wide-column store (Cassandra/ScyllaDB/Bigtable-style) fits: append-heavy, read-recent (SSTables and LSM-Trees).

> **Key term — partition key, clustering key, wide-column, LSM-tree:** In a wide-column database such as Cassandra, the **partition key** decides which machine stores a row, and all rows with the same partition key sit together. The **clustering key** decides the sort order of rows inside that partition, so "newest 50 messages" is one sequential read. An **LSM-tree** (log-structured merge-tree) is a storage engine that buffers writes in memory and flushes them to disk as sorted files that are merged later, which makes writes very cheap. Discord describes exactly this key shape (channel + time bucket) in its engineering posts on storing billions and later trillions of messages.

- **`conversations`** — `conv_id → type (1:1/group), members, last_seq, created_at`. Small, relational (SQL tables) or key-value (look up a value by its key).
- **`user_inbox`** (per-user change log) — partition `user_id`, clustering `inbox_seq`: rows like `(conv_id, seq)` meaning "conversation X has a new message". A device's sync cursor is just its last `inbox_seq`. This is the per-user read model (a copy of data shaped for one kind of query) that makes multi-device sync one range scan (read a run of consecutive rows in key order).
- **`read_state`** — `(user_id, conv_id) → last_delivered_seq, last_read_seq`. Unread count = `conv.last_seq − last_read_seq`, no counting needed.
- **`sessions`** (in-memory, TTL = time to live: entries delete themselves after a set time) — `user_id → [(device_id, gateway_id)]`. Written on connect, expired by missing heartbeats.

5. The architecture, annotated



Walking one message from Alice to Bob:

1. Alice's phone sends `send{client_msg_id=u1, conv_id=c9, body}` to **Gateway A**. Gateways are the "hold sockets, own nothing" tier from Bidirectional Streaming; they answer heartbeats locally.
2. Gateway A forwards to the **chat service instance that owns `c9`**. Ownership is assigned by consistent hashing (a way to map keys to servers so that adding or removing a server moves only a small share of keys) on `conv_id`, so exactly one instance is the **sequencer** for each conversation at a time (the "ownership over agreement" thread from Module 12).
3. The owner checks `u1` against recently seen `client_msg_id`s (dedupe), assigns `seq = last_seq + 1`, and writes the message to the **message store** and a pointer row to each member's **inbox log**. Only after these writes succeed does it send `ack{u1, seq=812}` back to Alice. The ack is the durability promise.
4. It looks up Bob's devices in the **session registry**, and publishes the message to the gateways holding them (Gateway B) via the **routing backplane** (an internal message bus connecting the backend to all gateways) (Publish-Subscribe: senders publish to a topic, and everyone subscribed to it receives the message). Gateway B writes it to both of Bob's sockets.
5. If Bob has no live session, the **push service** sends a notification through Apple/Google push (the notification capstone covers that pipeline). When Bob opens the app, it calls `GET /sync?cursor=…` and receives everything after its cursor.

Per-user vs per-conversation routing. Module 02 routed chat through *one topic per room*. That is ideal for big channels where most members are online. For 500M users in mostly 1:1 chats, one topic per conversation would mean billions of topics; routing by **user → gateway** (session registry) is cheaper. Real systems mix both: per-user routing for DMs and small groups, per-channel topics for large, busy rooms.

6. Deep dives

Deep dive 1 — ordering and "exactly-once" display without distributed locks. (A *distributed lock* is a lock that many machines coordinate over the network; slow and fragile, so best avoided.)

- *The problem:* two members of a group send at the same instant from different gateways. If each gateway stamped its own time, clocks that disagree by a few milliseconds would give different members different orders (Unreliable Clocks).
- *The fix:* **one sequencer per conversation** assigns a gap-free, increasing `seq`. Order is defined by the sequencer, not by clocks. This is DDIA's sequence number ordering applied per conversation, and it is cheap because a single conversation's rate is small (even a busy group sends a few messages per second).
- *Duplicates:* the phone retries `send` when it doesn't see an ack (tunnel, timeout). The sequencer keeps the last few thousand `client_msg_id → seq` mappings per conversation; a retry

gets the **original ack back**, not a new message. Delivery to recipients is at-least-once (a gateway may resend after a reconnect), and recipients drop anything with a `seq` they already have. **Idempotent send + dedupe on seq = messages appear exactly once**, even though no network hop is exactly-once (Exactly-Once Semantics makes the same argument).

- *Gaps:* because `seq` is gap-free, a client that holds 810 and receives 812 knows 811 is missing and fetches it. Gap detection is free.

Deep dive 2 — offline catch-up and multi-device sync.

- Each device keeps **one cursor**: its last `inbox_seq`. On reconnect it calls `/sync`, receives the list of `(conv_id, seq)` pointers since the cursor (one partition range scan on `user_inbox`), and fetches the message bodies it lacks. Then it rejoins the live stream and drops overlaps by `seq`. This is the "resume" recipe from Module 02, made per-user so that one call covers every conversation.
- A phone offline for a week in 200 active groups might have 20K new messages. Return **conversation summaries first** (last message + unread count per conversation) and let the app fetch full history lazily when a chat is opened. That keeps reconnect under a second.
- **Read receipts are cumulative** ("read up to 812"), so a device reading 50 messages sends one frame, not 50. Receipts to the *sender* in large groups are aggregated ("read by 37") and throttled; per-member receipts in a 1,000-member group would multiply traffic 1,000×.
- Your *own* sent messages must appear on your other devices too, so the sender's own inbox log also gets a pointer row (read-your-writes across devices; see DDIA's reading your own writes).

Deep dive 3 — presence without a heartbeat storm.

- *Naive:* broadcast "Alice is online/offline" to all of Alice's contacts on every change. With 150M online users flapping on mobile networks, that is millions of fan-out events per second, mostly to people who are not looking.
- *Better:* presence lives in the in-memory **session registry** with a TTL (for example 60 s, refreshed by the gateway on heartbeats). Presence is **pulled** when you open a chat or contact list, and **pushed** only to clients that have that chat open (a subscription that the gateway drops when the screen closes).
- Debounce flapping: only publish "offline" after the TTL expires, so a 5-second tunnel doesn't produce two events. Large groups show no per-member presence at all.
- Presence is **ephemeral data**: never write it to durable storage; losing it means a few seconds of wrong dots, which is acceptable (Graceful Degradation).

7. The failure walk

1. **A gateway dies** (1/3,000 of connections, ~50K users): their sockets drop; clients reconnect to another gateway with **jittered backoff** (wait longer after each failed try, plus a random extra delay so thousands of clients don't retry at the same instant) (Retry with Exponential Backoff), re-register sessions, and `/sync` fills any gap. Nothing durable lived on the gateway, so nothing is lost. Deploys work the same way (drain = stop accepting new connections and let existing ones finish, send a "going away" frame, reconnect).
2. **A conversation owner (sequencer) dies:** the hash ring (consistent hashing's circle of servers) moves its conversations to a neighbour. The new owner reads `last_seq` from the message store before accepting writes. To prevent two owners during the handover (a paused old owner waking up), writes are **conditional on `seq` not existing** or fenced with an ownership epoch (a number that increases with each new owner, so storage rejects writes carrying an old number); see DDIA's truth is defined by the majority for fencing tokens. Senders see a ~1–2 s pause and their retries are deduped.

3. **Message store slow:** the owner cannot ack, so senders see a "sending…" clock icon. That is correct behaviour: **never ack before durable**. A circuit breaker (a switch that stops calling a failing dependency for a while, so it can recover) stops piling requests onto the store; the client keeps the message in its local outbox (a queue of unsent messages on the phone) and retries.
4. **Push provider (APNs/FCM: Apple Push Notification service and Firebase Cloud Messaging, the services that wake phones) outage:** users with the app closed don't get notified, but nothing is lost; the messages are in their inbox logs and appear on next open. Queue pushes with a short TTL; a notification about a two-hour-old message is noise.
5. **New Year's Eve, 5× normal traffic:** the gateway tier is pre-scaled on a calendar (Auto-Scaling); low-value traffic (typing indicators, presence pushes, per-member group receipts) is shed (deliberately dropped to save capacity) first, messages last.
6. **A hot group** (a 1,000-member group where everyone replies at once): one sequencer handles it serially, which is fine for hundreds of messages/s. Fan-out of each message to 1,000 members is the cost; batch deliveries per gateway (one backplane message per gateway, not per member).

8. Trade-offs

Decision	Chosen	Alternative	Why
Ordering	Per-conversation sequencer	Client or server timestamps	Clocks disagree; a single owner per conversation is cheap and gives gap detection
Delivery guarantee	At-least-once + dedupe by `client_msg_id` and `seq`	Exactly-once transport	No network is exactly-once; idempotency gives the same user-visible result
History storage	Wide-column LSM store, partition by `(conv, bucket)`	Relational DB	Append-heavy, range reads by recency, petabytes; joins (combining rows from several tables) not needed
Routing	Per-user session registry (+ topics for big channels)	One topic per conversation	Billions of mostly-idle 1:1 conversations
Server history	Stored server-side for multi-device	Store-and-forward then delete (WhatsApp's original model)	Multi-device sync needs a server copy; E2EE (end-to-end encrypted) variants encrypt it
Presence	Pull + subscribe while visible, TTL in memory	Broadcast every change	Presence churn would dominate traffic

9. The follow-up questions

- **"How would you add end-to-end encryption?"** The server then stores and routes **ciphertext** (encrypted bytes) it cannot read. The Signal Protocol (published by Signal; used by WhatsApp per its security whitepaper) establishes per-device keys (secret numbers used to encrypt and decrypt). Group messages use sender keys so one encryption serves many recipients. Consequences: no server-side search or content moderation, and each of a user's devices is a separate recipient (fan-out per device, not per user). *Trap:* claiming you can keep server-side search unchanged.
- **"How do you generate `msg_id`?"** A globally unique, roughly time-ordered 64-bit ID (Snowflake-style: timestamp + machine + counter) for storage and deduplication, *separate* from `seq`, which defines order inside a conversation. Don't use the ID's timestamp for ordering.
- **"A user is in 5,000 groups. Does the inbox log explode?"** Only active conversations write to it, and the log is trimmed after all of the user's devices have passed a point (plus a retention window). Very large broadcast channels (100K+ followers) switch to **pull**: members read the channel's log directly instead of receiving a pointer each (the push/pull hybrid from the news feed).

- **"Edit and delete?"** Append an `edit`/`delete` event with its own `seq` that refers to the original; clients apply it. History stays append-only and sync stays one cursor (Event Sourcing thinking: store the list of changes, not just the latest state).
- **"Search chat history?"** A per-user search index built asynchronously from the inbox log via CDC (change data capture: streaming every database change to other systems); partition it by user so every query hits one shard (a **shard** is one partition of a database living on its own server) (DDIA's document-partitioned indexes).

Pattern manifest

Pattern	Where it served
Bidirectional Streaming	One WebSocket per device, own protocol (acks, seq, heartbeats, resume)
Publish-Subscribe	Routing backplane between chat service and gateways
Consistent Hashing + ownership	One sequencer per conversation
Idempotency	`client_msg_id` on send; dedupe by `seq` on delivery
Sharding	Messages by `(conv_id, bucket)`, inbox by `user_id`
CQRS	Inbox log, unread counts and search as derived read models
Cursor-Based Pagination	`/sync` cursor and history scroll-back
Retry with Backoff + jitter	Client reconnect after gateway loss
Graceful Degradation	Shed typing/presence/receipts before messages
Auto-Scaling (calendar)	Gateway tier before known peaks

Interview one-liner: "Gateways hold sockets and own nothing; one sequencer per conversation assigns gap-free sequence numbers; messages are acked only after they're durable in an append-only log partitioned by conversation; delivery is at-least-once with dedupe by client ID and sequence; each device syncs with one cursor over a per-user inbox log."

TL;DR

- **Problem:** 150M open sockets, per-conversation order, no loss, several devices per user.
- **Mechanism:** gateway tier + session registry + backplane; a per-conversation sequencer; wide-column message log keyed `(conv, bucket)`; per-user inbox log with device cursors; push notifications for offline users.
- **Hard parts:** ordering without clocks (sequencer), duplicates (idempotency keys), reconnect (cursor sync), presence (pull + TTL, never broadcast).
- **Price:** stateful connection tier, petabytes of mostly-cold history, and a protocol you own.

Case 2 — Email Service

The prompt (in short). Build a large webmail provider like Gmail or Outlook.com: users send and receive mail with anyone on the internet, read it in web and mobile apps, organise it with folders or labels, search it, and get protection from spam and malware.

What makes it hard. Unlike chat, you don't control the other side. Email arrives from millions of foreign servers over **SMTP** (Simple Mail Transfer Protocol, RFC 5321; an **RFC** is one of the numbered documents that define internet standards), many of them hostile, and your outgoing mail goes to

servers that may be down for days or may decide you're a spammer. Inside, the problem is storage: every user keeps years of mail and expects instant search over all of it.

1. Requirements and numbers

Functional: receive mail from the internet; send mail to any address; web/mobile clients list, read, thread, label, archive and delete; full-text search (search inside the words of every message); attachments; spam and malware (malicious software) filtering; standard protocol access (IMAP, the protocol mail apps use to read mail kept on a server, and SMTP submission for sending) for third-party clients.

Non-functional: once we accept a message, we never lose it (this is the core promise of email); reading the inbox **p99 < 300 ms**; search **< 1 s**; sending is asynchronous but reliable (retries for days); strong consistency (every reader sees the latest write) *within one mailbox* (if I archive a message on my phone, my laptop must not show it in the inbox a minute later).

Numbers (assumptions):

- **1B users**. Each receives **40 legitimate messages/day** and sends **10**.
- Inbound attempts are much higher than legitimate mail: assume **as many spam attempts as real mail**, most rejected at connection time. Accepted inbound: **40B/day ≈ 460K/s average, ~1.4M/s peak**. Outbound: **10B/day ≈ 115K/s average**.
- Average stored message (headers + body + share of attachments, after compression) ≈ **30 KB** → 40B × 30 KB = **1.2 PB/day** of new mail, **~440 PB/year** before replication. Per user that's ~1.2 MB/day, ~440 MB/year: plausible against the multi-GB free quotas big providers give.
- Metadata per message (IDs, sender, subject, snippet, labels, pointers) ≈ **1 KB** → **40 TB/day**. Metadata is ~3% of bytes but ~100% of queries.
- Search index ≈ 20–30% of text size; built per mailbox.

What the numbers say: storage is enormous, but **every query is scoped to a single mailbox**. That one fact makes the user the natural shard key (the field that decides which shard holds a record) for almost everything.

2. The five-questions pass

- **Flow:** receiving: the remote server waits for our `250 OK` (SMTP's "accepted" reply code); everything after that (spam checks, indexing, notifications) is asynchronous. Sending: the user's click returns as soon as the message is queued; delivery to the internet happens in the background with retries.
- **Storage:** the source of truth is the **raw message (MIME blob: the email in its standard internet format, with attachments encoded inside) + its per-mailbox metadata row**. Threads, label views, unread counts and the search index are derived.
- **Speed:** "first page of inbox" and "open message" dominate reads; both are single-mailbox, recency-ordered.
- **Failure:** remote servers are slow, down, or malicious; queues with retries and backoff everywhere; spam filters may be unavailable.
- **Growth:** storage per user grows forever; a few mailboxes (mailing-list archives, ticket systems) receive millions of messages.
- *Skeleton:* an SMTP edge that accepts into a replicated spool (a disk-backed holding area for received messages, copied to several machines), an async processing pipeline, mailbox-sharded metadata + a blob store, per-mailbox search indexes, and an outbound queue that retries per destination.

3. API

Clients use a JSON-over-HTTPS API (JSON is a simple text format for structured data; modern designs resemble **JMAP**, the JSON Meta Application Protocol, RFC 8620/8621); third-party clients use IMAP (RFC

9051) and SMTP submission on port 587 (a port is a numbered door on a server; each protocol has its usual number).

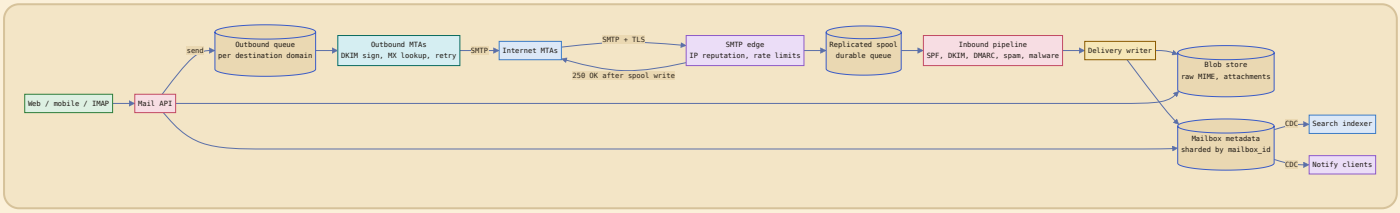
Endpoint	Purpose
`GET /threads?label=INBOX&cursor=…&limit=50`	Inbox page, newest first (cursor pagination)
`GET /messages/{id}` / `GET /messages/{id}/attachments/{n}`	Read a message; attachments stream from the blob store
`POST /messages/send` + `Idempotency-Key` header	Queue a message for sending; a double-click must not send twice (Idempotency)
`PATCH /messages/{id}` `{addLabels, removeLabels, read}`	Organise; label changes are the most frequent write
`GET /search?q=from:ana invoice after:2026/01/01`	Full-text + field search within the mailbox
`GET /changes?since=<state>`	Sync: "what changed since state X" (a per-mailbox cursor)

4. Data model

Everything is keyed by `mailbox_id` (≈ user) so each mailbox lives on one shard (Sharding).

- **`messages`** — `(mailbox_id, message_id)` → `thread_id, received_at, from, to, subject, snippet, size, label_bits, blob_ref, spam_score, modseq`. `label_bits` is a small bitset (a row of yes/no bits, one per label: INBOX, UNREAD, STARRED, user labels 1…n), so archiving is flipping a bit, not moving data.
- **`label_index`** — `(mailbox_id, label, received_at DESC, message_id)`: the read model behind "show INBOX newest first". Updated together with `messages` in one single-shard transaction (a group of writes that either all happen or none do).
- **`threads`** — `(mailbox_id, thread_id)` → `last_received_at, message_ids, label_union`. Threading uses the `Message-ID`, `In-Reply-To` and `References` headers defined in RFC 5322, falling back to subject + participants.
- **`changes`** — per-mailbox append-only log of `(modseq, message_id, what changed)`. Every write increments the mailbox's **modseq** (modification sequence; IMAP's CONDSTORE extension uses the same idea). Clients sync with "give me changes after modseq 91,204".
- **Blob store** — raw RFC 5322 message bytes, immutable (never changed once written), in object storage (a cheap, huge store of files addressed by key, such as Amazon S3). Attachments are stored **content-addressed** (key = SHA-256 of the bytes; SHA-256 is a hash function that turns any data into a 256-bit fingerprint, and identical data always gives the same fingerprint), so the same 5 MB PDF sent to 2,000 employees is stored once and referenced 2,000 times.
- **Search index** — one inverted index (a map from each word to the messages that contain it) per mailbox, co-located with (stored on the same machines as) the mailbox shard.

5. The architecture, annotated



- **SMTP edge:** terminates TLS (decrypts the encrypted connection; TLS is the encryption layer under HTTPS and secure SMTP), checks the connecting IP address against reputation lists (shared lists of known spam sources), applies per-IP rate limits, and rejects obvious junk **during the SMTP**

conversation (cheapest place to say no: we store nothing). It then writes the message to a **replicated spool** and only then answers `250 OK`.

- **Inbound pipeline** (consumers of the spool, a message queue): verifies sender authentication, scores spam, scans attachments for malware, applies the user's filters, and hands the result to the delivery writer. Spam and phishing (fake mail that tricks people into giving away passwords or money) classifiers are ML (machine learning) models served like any other model (Model Serving).
- **Delivery writer:** writes blob(s) first, then the metadata row and label index in one mailbox-shard transaction ("blob first, pointer last", so a crash never leaves a pointer to missing bytes). Idempotent on `(mailbox_id, message fingerprint)`.
- **Search indexer and notifications** consume the metadata change stream (Change Data Capture); an indexing delay of a few seconds is acceptable.
- **Outbound:** the API queues sent mail; outbound MTAs (mail transfer agents: servers that relay mail between domains) sign it, look up the recipient domain's MX records in DNS (DNS is the internet's name-to-address directory; an MX record names the servers that accept a domain's mail), and deliver with retries.

6. Deep dives

Deep dive 1 — the `250 OK` durability boundary.

- In SMTP, answering `250 OK` to the end of the message data means **"I have taken responsibility for this message"**; the sender deletes its copy. RFC 5321 is explicit that the receiver must not lose it afterwards. So the spool is our write-ahead log: the message must be on durable storage on **at least two machines (ideally two zones)** before we reply (a **zone**, or availability zone, is a separate data-centre building with its own power and network).
- What if we crash *after* writing but *before* the `250` reaches the sender? The sender times out and retries, and we receive the message twice. SMTP is therefore **at-least-once** by design. The delivery writer dedupes on a fingerprint (the `Message-ID` header plus a body hash, within a window of a few days; a **hash** is a short value computed from data that changes if the data changes). This is the same "idempotency under at-least-once" thread as everywhere else in the course.
- Why not deliver straight into the mailbox before answering? Because the pipeline (spam scoring, malware scanning) can take seconds, and SMTP senders time out. **Accept fast into a log, process asynchronously**: the Message Queue lesson's core idea.

Deep dive 2 — sending to an internet you don't control.

- *Authentication:* receiving servers now require it. The sender's domain publishes **SPF** (RFC 7208: which IPs may send for it), signs each message with **DKIM** (RFC 6376: a signature over headers and body, public key in DNS; the sender signs with a private key and anyone can check the signature with the matching public key), and publishes a **DMARC** policy (what receivers should do when the checks fail, and where to send reports; first described in RFC 7489, now standardised as RFC 9989, 2026). Google's and Yahoo's 2024 bulk-sender requirements made all three effectively mandatory for large senders. Our inbound pipeline runs the same checks on others.
- *Retries:* destination servers are often temporarily unavailable or reply with a `4xx` ("try later", often deliberate *greylisting*: refusing the first attempt from an unknown sender, because real servers retry and many spam tools don't). RFC 5321 says the give-up time should generally be **at least 4–5 days**, and that the retry interval should generally be **at least 30 minutes** (smarter, variable schedules are allowed when the client knows why delivery failed). So outbound is a durable queue with exponential backoff: for example a first quick retry after a few minutes for greylisting, then every 30–60 minutes, growing towards a few hours. After

the give-up time, or on a permanent `5xx` ("never" error), we send the user a **bounce** (a delivery status notification).

- *Per-destination backpressure* (slowing the sender down to the rate the receiver can take): a big receiver (say a large corporate domain) accepts only so many connections and messages per minute from one sender IP. Partition the outbound queue by **destination domain** (Partitioned Consumption) and give each domain its own concurrency limit. One slow or blocking domain then builds its own backlog instead of blocking everyone (Backpressure, Bulkhead: separate compartments so one failure can't sink the whole ship).
- *Reputation:* receivers judge us by IP and domain reputation. Keep separate IP pools for normal user mail and for anything risky (newly created accounts, bulk senders), and rate-limit new accounts: a spammer who signs up must not damage the IPs everyone else sends from.

Deep dive 3 — search inside one mailbox.

- A global inverted index over all mail ("term → every message containing it, for everyone") would force every query to filter by user and would be written by everyone at once. Instead, build **one small index per mailbox**, stored on the same shard as the mailbox. This is DDIA's document-partitioned secondary index: usually a scatter-gather problem (a query sent to every shard, with results merged), but here **every query carries the partition key** (the user), so it hits exactly one shard.
- Index asynchronously from the change stream. Newly arrived mail is searchable within seconds; a message that is not yet indexed can still be found through a short "recent messages" scan that the query merges in (so the user never sees "I just got it but search can't find it").
- Attachments (PDFs, documents) are text-extracted in the pipeline and indexed too; this is the expensive part and can lag more.

7. The failure walk

1. **A spool node dies:** replicated spool, so messages survive on the other replica; consumers resume from their last committed offset (the saved position up to which they had finished processing). Senders whose connections were cut mid-transfer retry (SMTP retries by design).
2. **The spam classifier is down:** failing *open* (deliver everything to the inbox) invites phishing; failing *closed* (reject) loses mail. Middle path: keep accepting into the spool (it is a buffer), and hold unscored messages for a short time; if the outage lasts, deliver with a cheaper rule-based filter and label them "not fully scanned" (Graceful Degradation). Attachments with executable content (runnable programs) stay quarantined (held aside, not delivered).
3. **A destination domain is down for two days:** its partition of the outbound queue grows; other domains are unaffected. The user gets a "still trying" notice after a few hours and a bounce after the give-up period.
4. **Our sending IPs land on a blocklist:** traffic to affected receivers moves to another IP pool while the cause (usually one compromised or spamming account) is found and suspended. Per-account outbound rate limits keep one account from doing this.
5. **A mailbox shard fails:** replicas with automatic failover (a replica is promoted to take over when the primary, the copy that accepts writes, dies) (Primary-Replica); inbound delivery for those mailboxes waits in the spool (lag, not loss). Reads fail over to replicas; label writes wait for the new primary.
6. **A hot mailbox** (a support address receiving 2M messages/day): it is one shard key, so it gets a per-mailbox delivery rate limit (a cap on how many messages per second are accepted) and a dedicated shard; beyond that, the owner is moved to an archiving product. See DDIA's hot spots.

8. Trade-offs

Decision	Chosen	Alternative	Why
Shard key	`mailbox_id`	Message ID or time	Every query is per mailbox; no cross-shard reads
Accept path	Spool then `250 OK`, process async	Process fully before `250`	SMTP timeouts; decouple acceptance from slow scanning
Labels	Bitset on the message + label index	Copy messages between folders	Archive/label = flip a bit; one copy of the data
Attachments	Content-addressed blobs (dedupe)	One blob per message	Same file sent to thousands of people stored once
Search	Per-mailbox index, async	Global index	Queries always carry the user; smaller, isolated indexes
Spam outage	Buffer, then degrade with labelling	Fail open or fail closed	Neither losing mail nor delivering phishing is acceptable

9. The follow-up questions

- **"How is this different from the chat system?"** Chat controls both ends and optimises for latency; email federates with untrusted strangers and optimises for **durability and deliverability**. Latency of seconds to minutes is fine; losing an accepted message is not.
- **"How do you store a 25 MB attachment sent to 5,000 people?"** Once, content-addressed, with a reference count (how many mailboxes still point at it) or periodic mark-and-sweep garbage collection (regularly find every blob still referenced and delete the rest); each mailbox row points to it. Encrypt at rest (encrypt the stored bytes on disk); deletion of the last reference makes it eligible for deletion.
- **"A user deletes a message: is it gone?"** Deletion flips a bit (Trash), then after 30 days the metadata row is removed and the blob's reference dropped. Backups and replicas age out on their own schedule; for legal erasure use per-user encryption keys and destroy the key (crypto-shredding, as in the Module 12 payments design).
- **"How do you stop your own users sending spam?"** New-account send limits, per-account outbound rate limits, outbound spam scoring (the same classifier, pointed outwards), and separate IP pools so abuse can't damage everyone's reputation.
- **"Push new mail to the phone?"** The metadata change stream feeds a notification service (Module 02 notification capstone). The phone then syncs with `/changes?since=modseq`; the push is only a hint.

Pattern manifest

Pattern	Where it served
Write-Ahead Log	The spool: durable before `250 OK`
Message Queue	Inbound pipeline and outbound delivery
Idempotency	Dedupe of SMTP retries; send API idempotency key
Sharding	Everything keyed by mailbox
CQRS	Label index, threads, search index as read models
Change Data Capture	Metadata changes → search indexer, notifications
Retry with Backoff	Outbound delivery over days
Partitioned Consumption + Backpressure + Bulkhead	Outbound queue per destination domain
Rate Limiting	SMTP edge per IP; outbound per account
Graceful Degradation	Spam-filter outage handling
Primary-Replica	Mailbox shard failover

Interview one-liner: "Accept mail into a replicated spool before saying 250, process spam and indexing asynchronously, shard everything by mailbox so every query hits one shard, store raw messages and attachments as content-addressed blobs, and send through a per-domain outbound queue that authenticates with SPF/DKIM/DMARC and retries for days."

TL;DR

- **Problem:** never lose accepted mail, federate with an untrusted internet, and search years of mail per user.
- **Mechanism:** SMTP edge → replicated spool → async pipeline → blob store + mailbox-sharded metadata → per-mailbox search; outbound queue partitioned by destination with retries and authentication.
- **Hard parts:** the `250 OK` boundary (at-least-once + dedupe), deliverability (SPF/DKIM/DMARC, reputation, per-domain limits), per-mailbox search.
- **Price:** hundreds of petabytes per year, a spam arms race, and a reputation you can lose in an afternoon.

Case 3 — Video Platform (YouTube-like)

The prompt (in short). Build a video-sharing platform: creators upload videos of any length; the platform processes them into formats every device can play; viewers anywhere stream them smoothly on any connection, from a phone on 3G to a 4K TV.

What makes it hard. Two very different pipelines share one catalogue. **Upload and processing** is a heavy batch workload (large jobs run in the background, not while a user waits): each uploaded hour of video must be decoded and re-encoded into many versions. **Playback** is a staggering bandwidth

problem: the bytes leaving the system are measured in **terabits per second**, and popularity follows a power law (a few videos get most views; most videos are rarely watched).

1. Requirements and numbers

Functional: resumable upload of large files (an interrupted upload continues where it stopped); processing into multiple resolutions and codecs (a **codec** is the algorithm that compresses and decompresses video, such as H.264); thumbnails; playback with **adaptive bitrate** (the player switches quality as bandwidth changes); seeking; video metadata, view counts; basic moderation/copyright checks before publishing.

Non-functional: playback start **< 2 s**, **rebuffering < 0.5%** of watch time (rebuffering = the video freezing while it waits for more data); upload never has to restart from zero; a newly uploaded video is watchable within **minutes**; uploaded originals are never lost; cost per view as low as possible.

Numbers (assumptions):

- **Viewing:** 500M DAU × 5 videos × 6 min average = **15B minutes watched/day**. Average concurrency = 15B ÷ 1,440 min/day ≈ **10M streams**, peak ~2× = **20M streams**. At an average of **3 Mbps** (megabits per second) → **~60 Tbps peak egress**, **~340 PB/day** delivered. No single data centre can do this: the design is a **CDN problem** first (CDN).
- **Uploads:** YouTube has publicly said (first in 2019, and repeated on its official blog since) that more than 500 hours of video are uploaded every minute; use that: **500 h/min = 720K hours/day**, about **4M uploads/day ≈ 50 uploads/s** (average 10 min each).
- **Storage:** originals at ~10 Mbps ≈ 4.5 GB/hour → **3.2 PB/day**. An H.264 ladder (the set of versions at different qualities) of five renditions (one rendition = one encoded copy at a given resolution and bitrate) totals ~10 Mbps → another **~3.4 PB/day**. Roughly **2.4 EB/year** (exabytes; 1 EB = 1,000 PB) before replication, which is why cold, rarely watched videos move to cheaper, erasure-coded storage tiers (erasure coding stores data plus a few parity pieces from which lost pieces can be rebuilt, using far less space than full copies).
- **Encode compute (assumption):** **~10 CPU-core-hours** (one processor core busy for ten hours) to produce the H.264 ladder for one hour of video → 720K × 10 = **7.2M core-hours/day ≈ 300K cores busy around the clock**. More efficient codecs (VP9, AV1: newer codecs that give the same quality in fewer bytes) cost several times more CPU per hour, so they can't be applied to everything.
- **Read:write** (ratio of reads to writes) is extreme: each video is encoded once and may be viewed millions of times.

What the numbers say: egress dominates cost and design (edge caching); encode compute is the second bill (be selective about expensive codecs); the metadata database is small by comparison.

2. The five-questions pass

- **Flow:** upload is request-response for each chunk; everything after "upload complete" (transcode, thumbnails, checks, publish) is an **asynchronous workflow**; the creator polls or gets a notification when ready. Playback is plain HTTP GETs of small files from the nearest CDN edge.
- **Storage:** originals and encoded segments are immutable blobs in object storage; the metadata DB holds titles, owners, status, and the list of renditions. View counts are a separate, approximate pipeline.
- **Speed:** the hot read is "next 4-second segment of a popular video", served from an edge cache. Metadata pages are cached too (Cache-Aside: the app checks the cache first and, on a miss, loads from the database and fills the cache).
- **Failure:** a transcode worker dies mid-task → retry that task only; an edge fails → the player's next request goes elsewhere; the origin (the source servers the CDN fetches from) must survive cache-miss storms (floods of requests for something no cache holds).

- **Growth:** egress grows with watch time; encode compute grows with upload hours; storage grows forever.
- *Skeleton:* resumable chunked upload → object store → a transcoding DAG (directed acyclic graph: a set of steps with "must run before" arrows and no loops) on a worker pool → packaged segments (short slices of video) + manifests (playlists listing the segments) → origin → CDN → adaptive-bitrate players.

3. API

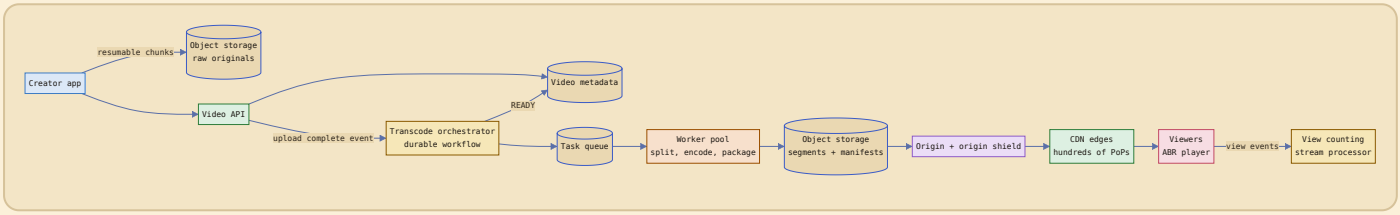
Endpoint	Purpose
`POST /videos {title, description, visibility}` → `{video_id, upload_url}`	Create the record (status `UPLOADING`) and hand out a resumable upload session
`PUT {upload_url}` with `Content-Range` (header saying which bytes this is) (or pre-signed multipart part URLs)	Upload in chunks of e.g. 8–16 MB; a dropped connection resumes from the last acknowledged byte. The open **tus** protocol and S3 multipart upload both work this way
`POST /videos/{id}/complete`	Upload finished; starts processing (idempotent)
`GET /videos/{id}`	Metadata, status (`PROCESSING` / `READY` / `FAILED`), and the **manifest URL**
`GET cdn…/{id}/master.m3u8` (HLS, Apple's streaming format) or `.mpd` (DASH, the open standard)	The manifest: list of renditions and their segments
`GET cdn…/{id}/720p/seg_00042.m4s`	One media segment; immutable, cacheable for a year
`POST /videos/{id}/views` (batched from the player)	Watch-time and view events for counting and recommendations

Uploads go **directly from the client to object storage** with pre-signed URLs; the API servers never carry the video bytes.

4. Data model

- **`videos`** (relational or key-value, sharded by `video_id`): `video_id, owner_id, title, description, status, duration, visibility, created_at, original_blob, manifest_path`.
- **`renditions`**: `(video_id, codec, height) → bitrate, segment_count, storage_path`. Lets the platform add an AV1 ladder later without touching anything else.
- **`processing_tasks`**: `(video_id, task_id) → type, chunk_index, rendition, state, attempts, output_ref`. The workflow's durable state.
- **Object storage layout:** `raw/{video_id}` (original), `enc/{video_id}/{codec}_{height}p/seg_{n}.m4s`, `enc/{video_id}/master.m3u8`, `thumbs/{video_id}/{n}.jpg`. Segments are **write-once**, so any cache anywhere can keep them forever.
- **View counters**: not a column that every view updates (a hot row, one record hammered by huge numbers of writes, for viral videos). Views flow as events into a stream processor (a program that handles an endless flow of events as they arrive) that dedupes and aggregates per minute, then updates a counter store (Stream Processing).

5. The architecture, annotated



1. The creator app creates the video and uploads the original in chunks **straight to object storage**, resuming after drops.
2. On completion, an event starts a **durable workflow** in the orchestrator (a Saga-style state machine, i.e. a process with named states and allowed moves between them, whose every step is recorded before it runs).
3. The orchestrator breaks the job into many small tasks on a queue; a large, auto-scaled worker pool runs them.
4. Workers write segments and manifests; when the required renditions exist, the video flips to `READY`.
5. Players fetch the manifest and segments from the nearest **CDN edge**; misses (requests the cache can't answer) go to a regional **origin shield** (one mid-tier cache per region that collapses edge misses) and only then to the origin.

6. Deep dives

Deep dive 1 — the transcoding pipeline as a parallel DAG.

- *Serial is too slow:* at ~10 core-hours per video-hour, one machine would take hours to process a 1-hour upload. Creators expect minutes.
- *Split by time:* cut the original into **~10-second chunks at keyframe boundaries** (a keyframe, or I-frame, is a frame that can be decoded on its own; cutting there means each chunk can be encoded independently). A 1-hour video → 360 chunks × 5 renditions = **1,800 independent tasks** of ~20 core-seconds each. With enough workers, encoding takes a couple of minutes, plus a final step that **stitches and packages** segments into HLS/DASH. Facebook describes this split-encode-stitch design in its SOSP 2017 paper (SOSP is a leading systems-research conference) on SVE (Streaming Video Engine), where processing starts while the upload is still arriving.
- *The DAG:* `validate → split → [encode chunk × rendition] → package → thumbnails → content checks → publish`. Content checks (copyright matching, safety classifiers) run in parallel with encoding, and `publish` waits for both.
- *Reliability:* each task is **idempotent**, keyed by `(video_id, chunk, rendition)`, and writes its output to a deterministic path; a retried task simply overwrites the same object (Idempotency). Workers take tasks with a lease (a claim that expires unless renewed); if a worker dies, the lease expires and another takes it. A task that fails repeatedly (a corrupt chunk) goes to a dead letter queue (a side queue for items that keep failing, kept for inspection) and the video is marked `FAILED` with a reason, instead of retrying forever.
- *Publish early, improve later:* make the video watchable as soon as the low and medium renditions exist; add 1080p/4K minutes later. Only videos that become **popular** get the expensive VP9/AV1 encodes: spending 5× the CPU is worth it only when the bandwidth saving is multiplied by millions of views. Netflix's published per-title encoding work applies the same logic to choosing bitrates per title.

Deep dive 2 — adaptive bitrate streaming (ABR).

- The video is not one file. Each rendition is cut into **segments of 2–6 seconds**, and a **manifest** lists them (HLS: RFC 8216; MPEG-DASH: ISO/IEC 23009-1). Segments of all renditions are aligned on the same timestamps, so the player can switch between renditions at any segment boundary.
- The player measures download throughput (data per second actually received) and watches its **buffer** (seconds of video downloaded but not yet played). Plenty of buffer and fast downloads → step up; buffer draining → step down before it runs dry. This is **client-side backpressure**: the consumer picks the rate it can sustain.
- An example ladder (assumed numbers, H.264):

Rendition	Bitrate	Typical use
240p	0.4 Mbps	Weak mobile signal
360p	0.8 Mbps	Mobile default on cellular
480p	1.4 Mbps	Small screens
720p	2.8 Mbps	Laptops
1080p	5 Mbps	TVs, good broadband
2160p (4K)	16 Mbps	Large TVs, popular content only

- Start-up: the player begins with a low rendition so the first frame appears within ~1–2 s, then climbs. Seeking just means fetching a different segment number: no server-side state per viewer at all, which is why plain HTTP caches can serve video.

Deep dive 3 — the CDN and the long tail.

- Segments are immutable, so they get long cache lifetimes; **manifests** of finished videos are immutable too. A hit at the edge costs a few milliseconds and no origin bandwidth.
- Popularity is skewed: the top few percent of videos account for most watch time. Those are almost always in edge caches. The **long tail** (millions of videos watched a few times a month) misses at the edge. Tiered caching (caches in layers, each asked only when the one below misses) handles it: edge → regional shield → origin. The shield turns 200 edge misses for the same segment into one origin read (Cache Stampede Prevention at CDN scale: request collapsing, i.e. many identical requests merged into one upstream fetch).
- Big platforms also **push** content: Netflix's Open Connect programme places its own cache servers inside ISPs (internet service providers, the companies that connect homes to the internet) and fills them with predicted-popular titles during off-peak hours, so peak-time traffic never crosses the backbone (the long-distance core links of the internet).
- Cost lever: every 1% of edge hit rate (share of requests answered from cache) on 340 PB/day is ~3.4 PB/day less origin traffic.

7. The failure walk

1. **Upload connection drops at 80%:** the client asks the upload session for the last committed offset and resumes. Incomplete sessions expire after a day and their parts are garbage-collected.
2. **A transcode worker dies:** its task lease expires; another worker re-runs just that one small task (~20 core-seconds), overwriting the same output path. The creator sees a slightly later "ready".
3. **The orchestrator crashes:** its state is in `processing_tasks` (durable), so a new instance resumes the DAG. No re-encoding of finished chunks.
4. **A viral video from an unknown creator:** cold at every edge at once. Shield-level request collapsing protects the origin; the popularity signal triggers pre-warming (loading into caches before anyone asks) to more PoPs (points of presence, i.e. CDN sites) and queues the efficient-codec re-encode.
5. **A CDN PoP (point of presence) fails:** DNS or anycast (one IP address announced from many sites, so the network routes each user to a nearby one) steers players to the next PoP; players simply retry the segment, maybe dropping a rendition for a few seconds. Very large platforms use **several CDNs** and switch between them by measured performance.
6. **Origin storage region outage:** originals and encoded segments are replicated across regions; the shield tier fails over. Uploads queue client-side and retry (Retry with Backoff).
7. **View counter lag:** counts shown may be minutes stale ("approximately 1.2M views"). Anti-fraud filtering happens in the stream job before counts are published.

8. Trade-offs

Decision	Chosen	Alternative	Why
Upload path	Client → object storage via pre-signed, resumable parts	Through API servers	Keeps multi-GB bodies off app servers; resumable
Encoding	Chunked, parallel DAG	One job per video	Minutes, not hours; retries are small
Codec strategy	Cheap ladder for all, efficient codecs for popular	Best codec for everything	CPU cost vs bandwidth saved scales with views
Delivery	HTTP segments + ABR via CDN	Custom streaming servers per viewer	Stateless, cacheable, works with any HTTP cache
Segment length	4–6 s (VOD)	1–2 s	Fewer requests and better compression; latency is not critical for VOD (video on demand, i.e. pre-recorded)
View counts	Approximate, streamed	Exact counter update per view	Avoid a hot row; tolerate minutes of staleness

9. The follow-up questions

- **"How do you stop someone downloading paid videos?"** Encrypt segments (e.g., CENC/AES, the common encryption scheme for video) and hand keys only through a DRM (digital rights management) licence server after authorisation; use signed, expiring CDN URLs or cookies. You raise the cost of copying; you can't make it impossible.
- **"How do you count views correctly?"** Define a view (e.g., ≥ 30 s watched), dedupe per session in a stream processor window, filter bots, and publish counts with a lag. Exact per-event counting in the metadata row would make every viral video a hot key.
- **"Why not make every segment 1 second for faster start?"** Many more requests per viewer (CDN request cost and overhead), worse compression (each segment starts with a keyframe), more manifest churn. Start-up is better solved with a low starting rendition and a well-placed edge.
- **"How do thumbnails and previews work?"** A task in the same DAG extracts frames at intervals; hover previews use a sprite sheet (many small frames in one image) so the player fetches one file.
- **"How would you add live streaming?"** The same segments, manifests and CDN, but the pipeline runs continuously in real time with tight latency budgets: that is Case 5.

Pattern manifest

Pattern	Where it served
CDN (tiered: edge → shield → origin)	All playback; request collapsing on misses
Message Queue + Auto-Scaling	Transcode task queue and worker pool
Saga / durable workflow	The processing DAG with recorded steps
Idempotency	Deterministic task outputs; `complete` call
Dead Letter Queue	Corrupt chunks don't block the pipeline
Backpressure (client-side)	ABR player picks the rendition it can sustain
Cache-Aside	Metadata and watch pages
Stream Processing	View counting and anti-fraud
Retry with Backoff	Resumable upload, segment retries
Sharding	Metadata by `video_id`

Interview one-liner: "Upload resumably straight to object storage, transcode as a parallel DAG of small idempotent chunk tasks, publish immutable HLS/DASH segments at several bitrates, and let adaptive players pull them through a tiered CDN; spend expensive codecs only where views repay them."

TL;DR

- **Problem:** encode once at high CPU cost, then deliver tens of Tbps to a power-law audience.
- **Mechanism:** resumable uploads → chunked parallel transcoding workflow → segments + manifests in object storage → origin shield → CDN edges → ABR players.
- **Hard parts:** encode latency and reliability (split, retry per task), smooth playback (ABR), egress cost and the long tail (tiered caching, push to ISPs, selective codecs).
- **Price:** exabytes of storage, hundreds of thousands of cores, and a CDN bill that dwarfs everything else.

Case 4 — File Sync and Storage (Google Drive / Dropbox-like)

The prompt (in short). Build a cloud drive: users keep a folder that is identical on their laptop, phone and the web; they can edit offline, share folders with others, recover old versions, and store large files.

What makes it hard. It looks like "upload files to storage", but it is really a **replicated database of file trees** where every device is a replica that can go offline for days and then come back with its own changes. The system must upload only what changed (not a whole 2 GB file for a one-line edit), keep every device converging on the same tree, and **never silently throw away** somebody's edit.

1. Requirements and numbers

Functional: upload/download files of any size; automatic sync across a user's devices; offline edits that sync later; shared folders with other users; version history and restore; trash; a change feed for the web and mobile apps.

Non-functional: durability above all (losing a file is the unforgivable failure); changes appear on other online devices within **~10 s**; bandwidth-efficient (only changed parts move); conflicts are **visible, never silently resolved** by discarding data; scales to multi-GB files and folders with hundreds of thousands of files.

Numbers (assumptions):

- **500M registered users, 100M DAU, ~3 devices each** → ~300M devices, most holding an idle notification connection.
- Stored data: **5 GB average per user** → **2.5 EB logical**. Deduplication (identical blocks stored once) saves perhaps **20–30%**.
- **2,000 files per user** → **1 trillion file entries**; at ~500 bytes per entry, **~500 TB of metadata** (plus revisions). Metadata is small next to the blobs but is where all the logic and all the queries are.
- Writes: each DAU saves **20 files/day** → **2B commits/day ≈ 23K/s average, ~70K/s peak**. With chunk-level dedupe, the average save uploads **~100 KB** of new data → **~200 TB/day** of new blocks.
- Change fan-out: each commit must notify the owner's other devices and every device of every member of a shared folder: typically ~3, occasionally tens of thousands.

What the numbers say: blob storage is a solved problem (object storage); the design is about the **metadata and sync protocol**: 70K atomic commits/s (each commit fully happens or not at all) against a trillion-entry namespace, and a change feed for 300M devices.

2. The five-questions pass

- **Flow:** the client uploads missing blocks, then makes one synchronous **commit** call; other devices learn about it asynchronously (a notification that says "something changed", followed by a pull of the changes).
- **Storage:** two stores with a strict order. **Blocks** (immutable, content-addressed bytes) and **metadata** (the tree, revisions, and a per-namespace change journal). The metadata journal is the source of truth for "what the folder looks like".
- **Speed:** the hottest operations are "anything changed since my cursor?" (answered from the journal) and "do you already have these block hashes?" (a key lookup).
- **Failure:** devices go offline mid-upload, two devices edit the same file, a client has a bug and tries to delete everything. Every write is a conditional commit against a known revision.
- **Growth:** storage grows forever; metadata grows with file count; hot spots are huge shared folders (a whole company in one namespace).
- *Skeleton:* client-side chunking and hashing → block service over object storage → metadata service with per-namespace journals → notification service → clients pull changes with a cursor.

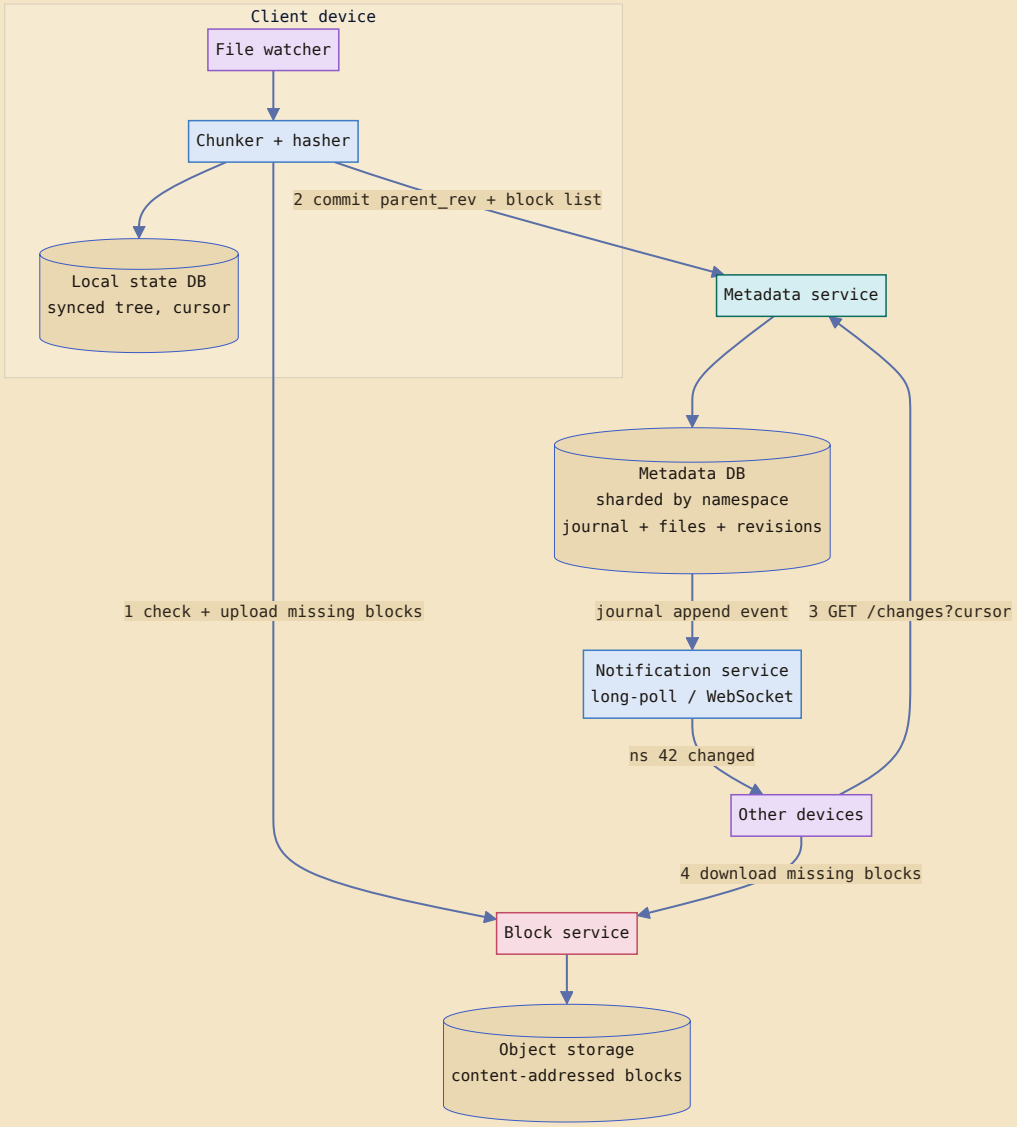
3. API

Endpoint	Purpose
`POST /blocks/check {hashes: [...]}` → `{missing: [...]}`	Which of my blocks does the server not have yet?
`PUT /blocks/{sha256}` (to a block server, often via pre-signed URL)	Upload one block; the server recomputes the hash and rejects mismatches
`POST /commit {ns_id, path, parent_rev, blocks: [...], size, mtime}`	Atomically create a new revision; `409 Conflict` (HTTP error: your base version is out of date) if `parent_rev` is no longer current
`GET /changes?cursor=…`	Journal entries after the cursor, paginated (cursor pagination); returns a new cursor
`GET /changes/wait?cursor=…` (long-poll, ~60 s: the server holds the request open until there is news)	Returns early as soon as anything changes; the client then calls `/changes`. Dropbox's public API has a long-poll endpoint of this kind, and Google Drive's API offers `changes.list` with a page token plus push "watch" channels
`GET /files/{id}?rev=…` → block list	Download: fetch blocks (from a CDN or block servers) and reassemble
`POST /shares {ns_id, member, role}`	Share a folder

4. Data model

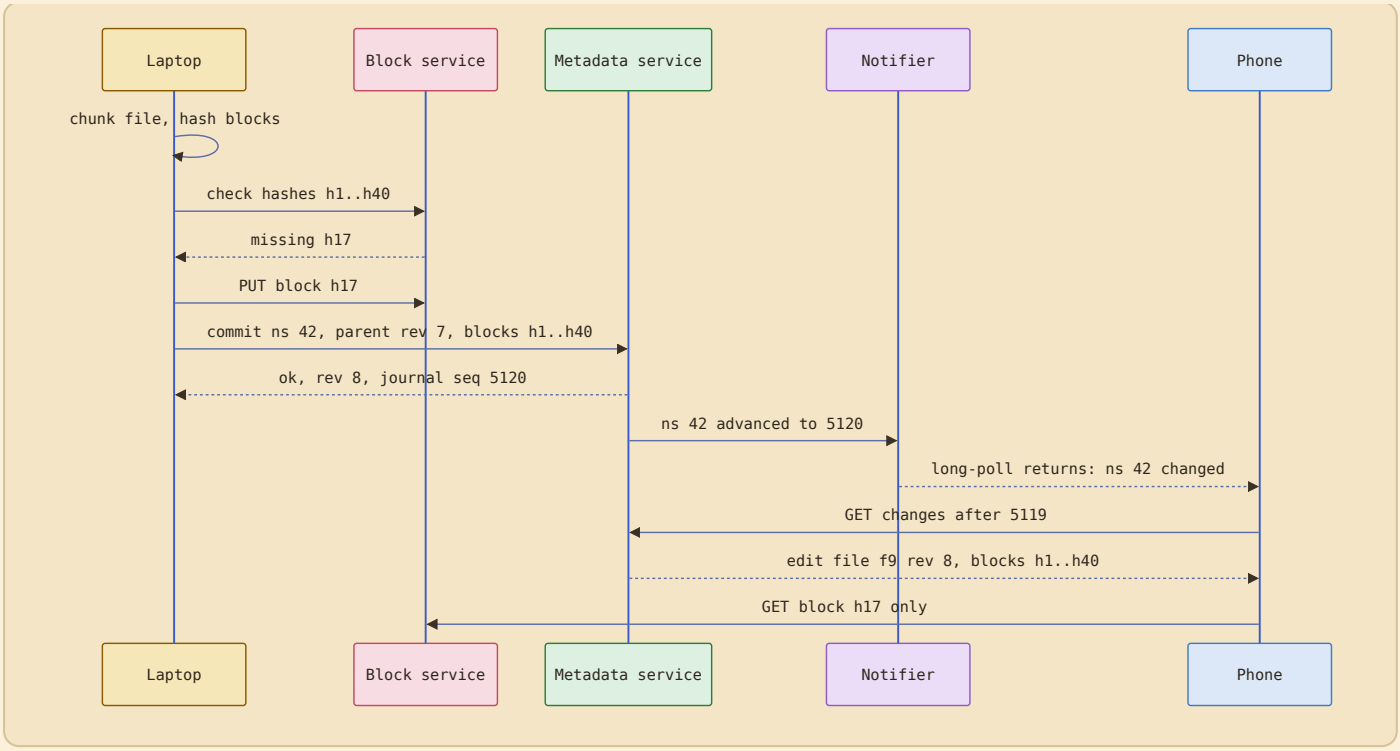
- **Namespace:** a unit of sharing and of ordering. Each user has a root namespace; **each shared folder is its own namespace** that is "mounted" into every member's tree. All metadata is sharded by `ns_id`, so every commit touches exactly one shard (Sharding).
- **`journal`** — `(ns_id, seq) → file_id, op (add/edit/move/delete), rev, path, block_list_ref, author_device, ts`. Append-only, gap-free `seq` per namespace. A device's **cursor** is a small map `{ns_id → last seq seen}`.
- **`files`** — `(ns_id, file_id) → parent_id, name, current_rev, size, is_deleted`. The current tree, derived from the journal (a read model kept in the same transaction).
- **`revisions`** — `(file_id, rev) → block hashes[], size, mtime, author`. Version history = older rows; restore = a new commit that points at an old block list (no data copied).
- **`blocks`** — `sha256 → storage location, size, refcount` (or no refcount; see garbage collection below). The bytes live in object storage, encrypted at rest.
- The metadata store needs transactions within a shard (check `parent_rev`, append to the journal, update `files`), so a sharded relational database (SQL tables split across many servers) is a natural fit.

5. The architecture, annotated



The order of the numbered steps is the key rule: **blocks first, metadata last**. A commit is only accepted if every block it references already exists, so no device can ever see a file that points at missing bytes. If the client crashes after uploading blocks but before committing, the only leftovers are unreferenced blocks, which garbage collection removes later.

One save, end to end:



6. Deep dives

Deep dive 1 — chunking, deduplication and delta sync.

- Files are split into **blocks**, and each block is named by the SHA-256 hash of its contents (**content addressing**). Two identical blocks anywhere have the same name, so they are stored once, and a client only uploads blocks the server reports missing. Dropbox has described storing file data as blocks of up to 4 MB addressed by hash in its posts about its Magic Pocket storage system.
- **Fixed-size blocks have a weakness:** insert a few bytes near the start of a file and every later block boundary shifts, so every later block looks new. **Content-defined chunking** (CDC; not to be confused with change data capture) instead places a boundary wherever a rolling hash (a hash over a sliding window of bytes that can be updated cheaply as the window moves) of the last few dozen bytes matches a pattern. Boundaries depend on the content, so after an insertion they re-align almost immediately. The idea comes from the LBFS paper (Muthitacharoen, Chen and Mazières, SOSP 2001); rsync (the classic Unix file-copying tool; Tridgell and Mackerras, 1996) solves the related problem of finding shifted matches with a rolling checksum.
- A small simulation makes the difference concrete (`code/chunking.py`, run with `.venv-books/bin/python`): a random 1 MB file, 5 bytes inserted in the middle, chunks of ~4 KiB (scaled down from real multi-MB blocks so the effect is visible).

```
fixed 4 KiB              chunks= 245  new chunks to upload= 123  bytes to upload=500,293
content-defined ~4 KiB   chunks= 189  new chunks to upload=   1  bytes to upload=10,626
```

With fixed blocks, half the file (everything after the edit) must be re-uploaded; with CDC, one chunk.

- *Dedupe scope is a security decision.* If the server dedupes **across users** and tells the client "I already have this block", an attacker can test whether *anyone* has a given file (a known confidential document) by trying to upload it. Safer: dedupe within a user or organisation, or always upload and dedupe silently on the server side.

- *Garbage collection:* blocks are shared, so deleting a file cannot delete its blocks directly. Use reference counts updated transactionally, or a periodic **mark-and-sweep** (mark every block referenced by any live revision; delete unmarked blocks older than a grace period, so blocks uploaded for a commit that hasn't landed yet survive).

Deep dive 2 — the change journal, cursors and conflicts.

- Every namespace has a **gap-free journal**. A device syncs by asking for entries after its cursor; this is DDIA's log-based replication idea with the device as a follower. Because each namespace has a single ordered journal, all devices apply the same operations in the same order and converge.
- **Commits are conditional** on `parent_rev`: "make rev 8 of this file, based on rev 7". This is compare-and-set (write only if the value is still what I last saw), DDIA's answer to lost updates. If the laptop and the phone both edited rev 7 while offline, the first commit wins and becomes rev 8; the second gets `409 Conflict`.
- **What to do with the loser:** never discard it. The client commits its version as a sibling file ("report (Phone's conflicted copy 2026-09-28).docx"), the approach Dropbox uses. Automatic merging only makes sense for formats the system understands (a collaborative document editor uses operational transforms or CRDTs, two techniques for merging simultaneous edits automatically; that is a different product). Compare Module 07's Vector Clocks (per-device counters that reveal whether two edits happened without knowing about each other): detect concurrency, keep both siblings, let someone merge; DDIA covers the same choice in handling write conflicts.
- **Tree operations are the subtle part.** "Device A renames folder X while device B adds a file inside X" or "A deletes X while B moves a file into X" must never lose the file. Handle moves as operations on stable `file_id`s (not paths), apply them in journal order, and resolve "parent was deleted" by restoring the parent. Dropbox's engineering blog on rewriting its sync engine (Nucleus) describes modelling the client's state as three trees (remote, local, and last-synced) and deriving the operations from their differences; it is worth reading for how hard this gets.
- **Safety valves:** a client bug that deletes 50,000 files will be synced faithfully. Keep deletes recoverable (trash + version history) and flag unusually large delete batches for confirmation.

Deep dive 3 — notifications to 300M devices.

- The notification service does **not** send data, only a hint: "namespace 42 now at seq 5,120". The device then pulls from the journal with its cursor. Hints can be lost, duplicated or coalesced (merged into one) without harm, because the journal is the truth (the "transport isn't the truth" rule from Bidirectional Streaming).
- Idle devices hold a long-poll or WebSocket connection to a notification gateway. Each gateway subscribes to the namespaces its devices care about (Publish-Subscribe).
- **Hot namespace:** a company-wide shared folder with 20,000 members and constant edits. Coalesce hints (at most one per namespace per second per device), and have devices add **jitter** before pulling so 20,000 devices don't hit the same shard in the same millisecond (Cache Stampede Prevention instinct). Very large namespaces get dedicated metadata shards.

7. The failure walk

1. **Upload interrupted:** blocks already uploaded stay; on retry, `/blocks/check` reports only the rest as missing. Commit happens once, at the end.
2. **Commit request times out:** did it apply? The client retries with the **same commit ID** (idempotency key); the metadata service returns the original result instead of creating rev 9 (Idempotency).

3. **Metadata shard primary fails:** failover to a replica (Primary-Replica); commits to that namespace pause for seconds; clients retry with backoff. Journals must use synchronous replication (the write is confirmed only after replicas also have it) (or a consensus-replicated log, where a majority of nodes must agree on each entry), because an acknowledged commit that disappears would make devices diverge.
4. **Notification service down:** devices fall back to polling `/changes` every few minutes. Sync gets slower, not wrong (Graceful Degradation).
5. **Block storage region lost:** blocks are replicated or erasure-coded across zones and regions; a commit is only acknowledged after the blocks it references are durable.
6. **A device offline for three months:** its cursor may point to journal entries that were compacted away (old entries removed to save space). The server answers "cursor too old", and the client does a **full resync**: list the current tree, compare with local state by hash, and upload or download the differences (conflicted copies for anything edited on both sides).
7. **Hash collision fear** (two different inputs giving the same hash): with SHA-256, an accidental collision is not a realistic risk; the server still verifies each uploaded block's hash so a buggy or malicious client can't store bytes under someone else's name.

8. Trade-offs

Decision	Chosen	Alternative	Why
Storage unit	Content-addressed blocks	Whole files	Dedupe, resumable upload, delta sync
Chunking	Content-defined (or fixed + rsync-style deltas)	Plain fixed blocks	Insertions don't shift every later block
Consistency unit	Per-namespace journal with gap-free `seq`	Global ordering	One shard per commit; total order only where needed
Concurrent edits	Conditional commit + conflicted copy	Last-writer-wins (newest write silently replaces the others)	Never silently discard a user's work
Change propagation	Hint + pull with cursor	Push full changes	Hints may be lost; the journal is the truth
Dedupe scope	Per user/organisation	Global	Global dedupe leaks whether others have a file

9. The follow-up questions

- **"How is this different from S3?"** S3 (Amazon's object storage service) stores objects by key; it has no tree, no devices, no sync, no conflict handling. This system *uses* object storage for blocks and adds the metadata journal and sync protocol on top.
- **"How do you support a 50 GB file?"** It is just 12,500 blocks of 4 MB; upload them in parallel and resumably, then commit a block list. The commit's metadata row stays small because it points to a block-list object.
- **"How do you make sharing permission changes take effect?"** Permissions live on the namespace; removing a member removes the mount from their tree via a journal entry in their root namespace, and the block service checks access on every block download (signed short-lived URLs: links with a tamper-proof signature and an expiry time).
- **"What about real-time co-editing like Google Docs?"** Different product: character-level operations over a WebSocket, merged with operational transforms or CRDTs, stored as an operation log. File sync treats files as opaque blobs.
- **"How do you save bandwidth on a LAN (local area network, e.g. one office's network) with ten laptops syncing the same folder?"** Peers on the same network can fetch blocks from each other,

verified by hash, since the hash is the name. (Dropbox shipped a LAN sync feature on this principle.)

Pattern manifest

Pattern	Where it served
Sharding	Metadata by namespace
Write-Ahead Log / event log	Per-namespace journal as the source of truth
CQRS	`files` tree as a read model of the journal
Idempotency	Commit IDs; content-addressed blocks are naturally idempotent
Vector Clocks' rule (keep siblings)	Conflicted copies instead of last-writer-wins
Publish-Subscribe	Notification hints to device gateways
Cursor-Based Pagination	`/changes?cursor`
Cache Stampede Prevention (jitter, coalescing)	Hot shared namespaces
Graceful Degradation	Polling fallback when notifications are down
Primary-Replica	Metadata shard failover

Interview one-liner: "Split files into content-addressed blocks so only new blocks move; upload blocks first and commit metadata last with a conditional, per-namespace journal append; other devices get a hint and pull changes with a cursor; concurrent edits become conflicted copies, never silent overwrites."

TL;DR

- **Problem:** many devices, often offline, must converge on the same file tree without losing anyone's edits, and without re-uploading whole files.
- **Mechanism:** client chunking + hashing → block service → conditional commit into a per-namespace journal → hint via long-poll → pull with cursor → download missing blocks.
- **Hard parts:** content-defined chunking and dedupe (with its privacy catch), conflicts and tree operations, notification fan-out on huge shared folders.
- **Price:** a trillion-row metadata store, a sync engine on every client (the hardest code in the product), and garbage collection of shared blocks.

Case 5 — Live Streaming

The prompt (in short). Build live video streaming: two shapes on one platform. (a) A creator platform like Twitch: ~100K channels live at once, most with a handful of viewers. (b) A sports event like a cricket final: **one** stream watched by **20M+ people at the same moment**. Viewers should see the action within a few seconds of it happening.

What makes it hard. Case 3 had hours to prepare each video and a CDN full of cached segments. Live has **no preparation time**: every segment is created, encoded, packaged and distributed within a few seconds, and in the big-event case every viewer requests **the same brand-new file at the same instant**. The two shapes fail differently: (a) is a cost problem (100K real-time encoders), (b) is a

thundering-herd problem (a huge crowd of clients hitting the same thing at the same instant) (one hot object, tens of Tbps).

1. Requirements and numbers

Functional: broadcaster sends a stream from software or hardware encoders (programs or devices that compress live video); viewers watch in real time with adaptive quality; a short rewind window (DVR); the recording becomes a normal video afterwards; live chat alongside (Case 1); viewer counts.

Non-functional: glass-to-glass latency (camera to viewer's screen) of **~3–5 s** for interactive creator streams, up to ~10–20 s acceptable for mass sports if it buys smoothness; **rebuffering < 1%**; the big event must not fail (no second chance); a broadcaster's brief network drop must not end the stream.

Numbers (assumptions):

- **Creator shape:** 100K concurrent channels × 6 Mbps source = **600 Gbps of ingest** (ingest = the incoming streams from broadcasters). Real-time transcoding to a 4-rung ladder costs (assume) **~4 CPU cores per channel → 400K cores** if everyone got it. Most channels have fewer than 10 viewers, so full ladders only for channels above a viewer threshold; the rest get source quality plus one low rendition, or run on hardware encoders.
- **Event shape: 25M concurrent viewers**. That is realistic, not extreme: Disney+ Hotstar reported about 25 million concurrent viewers for a 2019 World Cup match and about **59 million** for the 2023 Cricket World Cup final. At an average **3 Mbps → 75 Tbps** for a single stream.
- **The herd:** with 2-second segments, each new segment is requested by up to 25M players within roughly the same 2 s: **~12M requests/s for a handful of URLs**, plus manifest refreshes at the same rate.
- **The "play" spike:** at the start of the match (and after rain delays or wickets), millions press play within a minute. The **API tier** (login, entitlement checks, i.e. "has this user paid for this?", manifest URLs) sees its peak here, not during steady viewing.

What the numbers say: the creator shape is priced by **transcode cost**; the event shape is decided by **edge fan-out of one hot object** and by **pre-scaling the control plane** (the services that manage and coordinate, as opposed to the data plane that moves the video bytes) for a known moment.

2. The five-questions pass

- **Flow:** broadcaster → ingest is a long-lived push stream; viewers pull segments over HTTP (so every CDN in the world can help). Control actions (start stream, get manifest URL) are request-response.
- **Storage:** live segments are **ephemeral** (kept for the DVR window, e.g., 2 hours); the archive is written to object storage for VOD. The source of truth for "who is live where" is a small session registry.
- **Speed:** latency is the product. Every stage gets a budget in milliseconds.
- **Failure:** the broadcaster's uplink (their upload connection) is the weakest link; a transcoder crash must not end the stream; the big event needs redundancy at every layer.
- **Growth:** creator side scales with channel count (encode cost); event side with concurrent viewers (egress + herd).
- *Skeleton:* ingest PoPs → per-channel real-time pipeline (transcode + package into short segments) → origin → origin shield → CDN edges with request collapsing → low-latency ABR players.

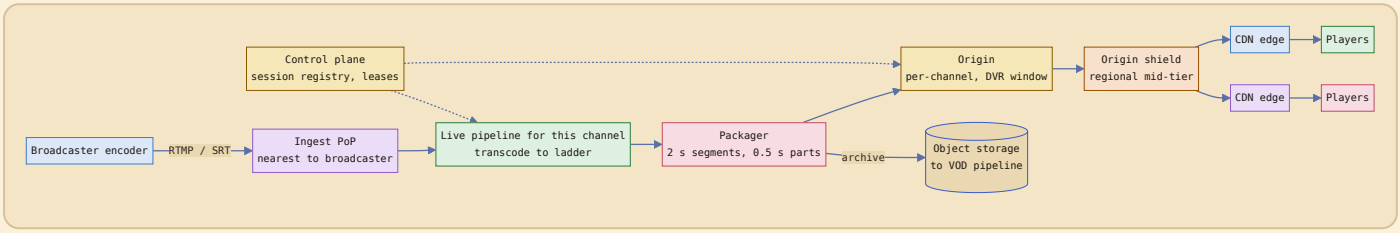
3. API

Interface	Purpose
`POST /channels/{id}/stream-key`	Create or rotate the secret key that authorises a broadcaster (store only its hash)
Ingest: `rtmp(s)://ingest.example.net/live/{stream_key}` or SRT / WHIP	Broadcaster pushes the stream. **RTMP** (Real-Time Messaging Protocol) is the long-standing default in encoder software; **SRT** (Secure Reliable Transport) adds packet recovery over lossy networks; **WHIP** (RFC 9725, 2025) is the IETF (internet standards body) standard for WebRTC-based ingest (WebRTC = the browser technology for real-time audio/video)
`GET /live/{channel}/master.m3u8` (or `.mpd`)	The live manifest; refreshed by the player every segment
`GET cdn…/{channel}/{rendition}/{segment}.m4s`	Media segments (and partial segments for low latency)
Webhooks (HTTP calls the platform makes to your server when something happens)/events `stream.online`, `stream.offline`	Tell followers, start/stop the archive, update listings
`POST /live/{channel}/heartbeat` (every 30–60 s, batched)	Viewer counts and quality metrics

4. Data model

- **`channels`** — `channel_id → owner, stream_key_hash, transcode_tier, latency_mode, settings`.
- **`live_sessions`** — `session_id → channel_id, started_at, ingest_pop, pipeline_id, status (starting/live/reconnecting/ended), renditions, archive_path`.
- **Pipeline assignment** — which transcoder node owns which live session, held as a **lease** in a coordination service (a small, highly reliable store such as ZooKeeper or etcd used to agree on who owns what; one owner at a time; see DDIA's membership and coordination services).
- **Segments** — on the packager/origin's local SSD (solid-state disk) or memory for the DVR window; named by **media timestamp** (e.g., `720p/t_000123456.m4s`), not by an arbitrary counter, so two redundant pipelines produce identical names for identical moments.
- **Viewer counts** — approximate, from player heartbeats aggregated in a stream processor per channel per 30 s.

5. The architecture, annotated



- **Ingest PoPs** close to broadcasters keep the fragile first mile short; the stream then travels on the provider's own backbone.
- The **live pipeline** for a channel is stateful (it holds the encoder state for that stream) and **pinned** to one transcoder node under a lease. It decodes once and encodes the ladder in real time (hardware encoders or GPUs, graphics processors good at parallel maths, make this cheaper).
- The **packager** emits short segments (e.g., 2 s) and, for low latency, **partial segments** (e.g., 0.5 s parts) (low-latency DASH sends them as chunked CMAF, a shared segment format usable by both HLS

and DASH; LL-HLS, low-latency HLS, lists them as "partial segments" in the playlist, which can be the same CMAF chunks, so one packager can serve both). It also writes the archive for Case 3's VOD pipeline.

- **Origin → shield → edges:** same tiers as Case 3, but with very different cache rules. Segments are immutable (long TTL); the **manifest changes every segment** (TTL of about half a segment, or the blocking-reload mechanism of LL-HLS where the player asks for "the next version" and the server holds the request until it exists).

6. Deep dives

Deep dive 1 — the latency budget. Where do the seconds go? An example budget for a ~4 s low-latency HTTP stream (assumed figures):

Stage	Time	Lever
Camera capture + broadcaster's encoder	0.3–1 s	Encoder settings (no long look-ahead)
First mile to ingest	0.1–0.5 s	Nearby ingest PoP; SRT for lossy links
Transcode	0.5–1 s	Hardware encoding; short GOPs (group of pictures: the frames from one keyframe to the next; here a keyframe every 2 s)
Packaging	0.5 s	Publish 0.5 s partial segments, not whole 2–6 s segments
Origin → shield → edge	0.1–0.3 s	Persistent connections; shields close to edges
Player buffer	1–2 s	The safety margin against network jitter (variation in packet arrival times)

- **Classic HLS** with 6 s segments and players buffering three segments adds up to **20–30 s**. Low-latency HLS/DASH cuts it to **2–5 s** by sending parts of a segment as soon as they exist. **WebRTC** reaches **< 1 s** but is stateful per viewer and not cacheable by ordinary CDNs, so it is used for interactive cases (auctions, co-streaming, video calls) rather than 25M-viewer sports.
- **The core trade:** a smaller player buffer means lower latency *and* more rebuffering on shaky networks. Mass sports often choose ~8–15 s for smoothness; creator streams where chat interacts with the host choose ~2–4 s. Make it a per-channel setting (`latency_mode`), not a platform-wide constant.

Deep dive 2 — fan-out of one hot object.

- 25M players ask for the same new segment within ~2 s. When it first appears, every edge misses at once; without protection each edge forwards thousands of identical requests to the shield, and each shield to the origin: a stampede on a file that is only milliseconds old.
- **Request collapsing** at every tier: the first request for a missing object goes upstream; every other request for the same object **waits on that one fetch** and is served from it. Facebook's 2015 engineering post on building Facebook Live describes this exact problem and fix at its edge and origin caches. It is Cache Stampede Prevention's single-flight (only one in-flight fetch per key; everyone else waits for its result), applied to a CDN. With collapsing, the origin serves each segment roughly **once per shield** instead of once per viewer.
- **Capacity is booked, not auto-scaled.** A 75 Tbps event exceeds what one CDN may have spare in a region, so big events use **multiple CDNs**, with players or a steering service splitting traffic between them. Everything is **pre-scaled from the event calendar** (Auto-Scaling: calendar first, reaction second): origin, shields, the API tier and the entitlement service.
- **The play-button storm:** protect login and entitlement with pre-scaling, rate limiting with a queue ("you're in line, starting in 20 s") rather than errors, and cache entitlement tokens so

a refresh doesn't re-check the database. Shed non-essential features first (recommendations, comments, high-resolution thumbnails) (Graceful Degradation).

- **Degrade quality, not availability:** if egress runs short, remove the top rendition (e.g., 1080p) from the manifest for some regions. Players step down smoothly; nobody's stream stops.

Deep dive 3 — keeping the stream alive.

- *Broadcaster drops for 5 s:* the session enters `reconnecting` instead of `ended`; the origin keeps the manifest alive (a grace window of e.g. 30–60 s), players wait on their buffer and then show "reconnecting"; when the encoder reconnects with the same key, the session resumes. Without the grace window, every blip would kick millions of viewers back to the home page.
- *Transcoder node dies:* the lease expires within a couple of seconds; the control plane starts the pipeline on another node, which re-attaches to the ingest. Viewers lose a few seconds (a stall or a skip), not the stream. Health checks (regular automatic "are you working?" probes) watch **segment output rate**, not just process liveness (Health Check Endpoint): a pipeline that is running but not producing segments is dead.
- *Premium events run 1+1:* two broadcaster encoders → two ingest PoPs in different regions → two independent pipelines. Because segments are named by media timestamp and encoder settings are identical, both pipelines produce **interchangeable segments**; the origin (or the player, via the redundant streams that HLS manifests can list) switches to the healthy one within a segment. Redundancy at every layer is the price of "no second chance".

7. The failure walk

1. **Broadcaster's network drops:** grace window, `reconnecting` state, resume on the same key (above).
2. **Transcoder crash:** lease expiry → new pipeline in seconds; for premium events, instant switch to the twin pipeline.
3. **Origin overloaded by misses:** request collapsing at edges and shields; origin sees one request per segment per shield. If a shield fails, edges fall back to another shield (more load, still collapsed).
4. **A CDN degrades during the event:** the steering service moves traffic to the other CDNs based on measured error rates and rebuffering from player telemetry (measurements players report back automatically). Players retry a failed segment on an alternate host.
5. **Play-button storm at kick-off:** pre-scaled API tier, waiting-room queue instead of errors, cached entitlements, non-essential features shed.
6. **Live chat on a 1M-viewer stream:** one room with a million members cannot deliver every message to everyone (a few thousand messages/s × 1M recipients). Apply slow mode (per-user send limits), show a sampled subset of messages, and aggregate reactions into counts (Backpressure: the chat stream drops by policy).
7. **Archive writer falls behind:** it is off the live path; live viewing is unaffected, and the VOD appears later.

8. Trade-offs

Decision	Chosen	Alternative	Why
Delivery protocol	LL-HLS/LL-DASH over CDNs	WebRTC to every viewer	HTTP is cacheable and scales to tens of millions; WebRTC only for sub-second interactive use
Segment size	2 s segments with 0.5 s parts	6 s segments	Latency of seconds instead of tens of seconds; costs more requests
Latency target	Per channel (2–4 s interactive, 8–15 s sports)	One global target	Latency trades directly against rebuffering
Transcoding	Full ladder only above a viewer threshold	Everything for everyone	Real-time encoding is the creator-shape bill
Big-event capacity	Pre-booked, multi-CDN, calendar pre-scaling	Reactive auto-scaling	A 10-minute scale-up is useless at kick-off
Redundancy	1+1 pipelines for premium events	Single pipeline + fast restart	"No second chance" events can't afford even a restart gap

9. The follow-up questions

- **"How does this relate to the video platform?"** Same segments, manifests, ABR players and CDN tiers; the difference is time. VOD has hours to encode and a warm cache; live has seconds and a cache that is always cold for the newest segment. The live recording flows into the VOD pipeline afterwards.
- **"Why not just use WebRTC everywhere for sub-second latency?"** Each viewer then needs a stateful media session through selective forwarding servers (servers that relay each sender's media to each viewer without re-encoding), which ordinary HTTP CDNs can't cache; cost and complexity rise steeply with audience size. Use it where interaction truly needs sub-second latency.
- **"How do you show an accurate viewer count?"** You don't need exact: count distinct players sending heartbeats in the last minute, aggregated per channel in a stream job; approximate distinct counting (e.g., HyperLogLog sketches: tiny data structures that estimate how many different items were seen, within about 1–2%) keeps it cheap at 25M.
- **"How do you insert ads into a live stream?"** Server-side ad insertion: at ad markers, the manifest for each viewer (or viewer segment) points to pre-encoded ad segments. That makes manifests personalised, so they are generated at the edge or by a manifest service while media segments stay shared and cacheable.
- **"How do you protect a paid event from restreaming?"** Signed, short-lived URLs and tokens per session, DRM for premium content, concurrent-session limits per account, and forensic watermarking (invisible per-viewer marks in the picture that identify who leaked a copy) for pirate take-downs. Expect leakage; limit it.

Pattern manifest

Pattern	Where it served
CDN (tiered) + request collapsing	The one-hot-object fan-out
Cache Stampede Prevention	Single-flight on new segments at edges and shields
Auto-Scaling (calendar first)	Pre-scaling for known events
Rate Limiting + waiting room	Play-button storm on the API tier
Graceful Degradation	Drop top renditions, shed non-essential features
Health Check Endpoint (output-based)	Detect pipelines that run but don't produce
Leases / ownership	One pipeline owner per live session
Backpressure (client-side ABR, chat drop-by-policy)	Players choose sustainable bitrates; chat sampling
Stream Processing	Viewer counts, quality telemetry, CDN steering
Bulkhead / redundancy	1+1 pipelines and multi-CDN for premium events

Interview one-liner: "Ingest near the broadcaster, run a pinned real-time pipeline per channel that emits short timestamp-named segments with partial chunks, and fan out through a tiered CDN with request collapsing; latency is a per-channel budget traded against rebuffering, and big events are pre-scaled, multi-CDN and fully redundant because there is no second chance."

TL;DR

- **Problem:** create and distribute each segment within seconds, cheaply for 100K small channels and safely for one 25M-viewer event.
- **Mechanism:** RTMP/SRT ingest PoPs → per-channel real-time transcode + packaging (LL-HLS/DASH) → origin → shields → edges → ABR players.
- **Hard parts:** the latency budget (segment/part size vs buffer), the hot-object herd (request collapsing, multi-CDN, pre-scaling), and survival (reconnect windows, leases, 1+1 pipelines).
- **Price:** real-time encode cost, booked capacity for peaks, and redundancy that sits idle most of the year.

Pattern Review

What this review is. One table showing where each course pattern showed up across the five cases, followed by the ideas that repeat. Cover a column and rebuild it from memory as a self-test.

Pattern (lesson)	Chat	Email	Video	File sync	Live
Bidirectional Streaming	Device WebSockets, own protocol	—	—	Notification long-poll / socket	Broadcaster ingest stream
Message Queue	—	Spool, outbound queue	Transcode tasks	—	Archive to VOD
Publish-Subscribe	Routing backplane	Change stream consumers	Upload-complete events	Namespace change hints	Stream online/offline events
Sharding	`(conv, bucket)`, inbox by user	By mailbox	Metadata by video	By namespace	Pipeline per channel
Consistent Hashing	Conversation owners	—	—	—	—
Write-Ahead Log / logs	Message log, inbox log	Spool before `250 OK`	Durable workflow state	Per-namespace journal	—
CQRS	Unread counts, conversation list	Label index, threads, search	—	`files` tree from journal	—
Cache Stampede Prevention	—	—	Shield request collapsing	Jittered pulls on hot folders	Collapsing on every new segment
Idempotency	`client_msg_id`, dedupe by `seq`	SMTP retry dedupe, send key	Deterministic task outputs	Commit IDs, content hashes	—
Retry with Backoff	Reconnect with jitter	Outbound delivery for days	Resumable upload	Commit retries	Segment retries on alternate host
Dead Letter Queue	—	Bounces after give-up	Corrupt chunks	—	—
Graceful Degradation	Shed typing/presence first	Spam-filter outage mode	Lower renditions	Polling fallback	Drop top rendition, waiting room
Auto-Scaling	Calendar peaks	—	Worker pool on queue depth	—	Event pre-scaling
Saga / durable workflow	—	—	Processing DAG	—	—
Vector Clocks' rule	—	—	—	Conflicted copies (keep siblings)	—
CDN	Media attachments	Attachments (optional)	All playback	Block downloads	All playback
Rate Limiting	Per-user send limits	SMTP edge, outbound per account	Upload quotas	API quotas	Play-button storm, chat slow mode
Cursor-Based Pagination	`/sync`, scroll-back	`/changes?since=modseq`	—	`/changes?cursor`	—

Pattern (lesson)	Chat	Email	Video	File sync	Live
Change Data Capture	Search index	Indexer, notifications	—	—	—
Stream Processing	—	—	View counts	—	Viewer counts, CDN steering
Backpressure	Capped socket buffers	Per-domain outbound limits	ABR player	—	ABR player, chat sampling

Ideas that repeat across all five

1. **Separate the bytes from the facts.** Every case keeps large immutable blobs (attachments, segments, file blocks) in object storage behind a CDN, and small mutable metadata in a sharded database. Blobs are written first and referenced last.
2. **A per-scope ordered log plus a cursor is how clients stay in sync.** Chat's per-conversation `seq` and per-user inbox, email's `modseq`, file sync's namespace journal: all are DDIA's partitioned log with the client as a follower. Order only within the scope that needs it.
3. **Hints are not data.** Push notifications, "namespace changed" pings, and presence dots may be lost or duplicated; correctness comes from pulling the log afterwards.
4. **At-least-once everywhere, made safe by idempotency.** SMTP retries, message resends, commit retries and re-run transcode tasks all repeat; client-generated IDs and content hashes absorb the repeats.
5. **The shard key is whatever every query carries.** Conversation, mailbox, video, namespace, channel. When a query always contains the key, even "secondary" indexes such as email search stay on one shard.
6. **Skew shows up again.** Big groups, hot mailboxes, viral videos, company-wide folders, one 25M-viewer stream: each needs its own escape (dedicated shards, request collapsing, pull instead of push, pre-booked capacity).
7. **Degrade the extras, protect the core.** Typing indicators, presence, 4K renditions and recommendations are shed long before a message, an email or the live picture.

Decision rules from this module

Choice	Rule
WebSocket vs long-poll vs push notification	Socket when the app is open and interactive; long-poll for idle sync hints; OS push (the phone operating system's notification service) only to wake a closed app
Per-user vs per-room routing	Per-user registry for DMs and small groups; per-room topics for large busy rooms; pull for huge broadcast channels
Accept then process vs process then accept	Accept into a durable log fast when the sender will time out (SMTP, uploads); process asynchronously
Fixed vs content-defined chunks	Content-defined when files are edited in place; fixed is fine for append-only or replaced-whole files
VOD vs live segment length	4–6 s for VOD (efficiency); 1–2 s with sub-second parts for live (latency)
HTTP streaming vs WebRTC	HTTP (HLS/DASH) for audiences; WebRTC only when sub-second interaction is the product
Cheap vs efficient codecs	Cheap for everything first; efficient codecs where views repay the CPU

Module TL;DR

- **Chat:** gateways hold sockets; one sequencer per conversation; append-only message log by `(conv, bucket)`; per-user inbox log with device cursors; at-least-once delivery with idempotent sends.
- **Email:** replicated spool before `250 OK`; async spam/malware pipeline; everything sharded by mailbox; content-addressed attachments; outbound queue per destination domain with SPF/DKIM/DMARC and days of retries.
- **Video:** resumable direct-to-storage uploads; chunked parallel transcoding DAG of idempotent tasks; immutable HLS/DASH segments at several bitrates; tiered CDN; expensive codecs only for popular videos.
- **File sync:** content-addressed (ideally content-defined) blocks; blocks first, conditional metadata commit last; per-namespace journal and cursors; hints then pull; conflicted copies instead of silent overwrites.
- **Live:** ingest PoPs → pinned real-time pipeline → short segments with partial chunks → shields and edges with request collapsing; latency is a per-channel budget; big events are pre-scaled, multi-CDN and 1+1 redundant.
- **Across all five:** separate blobs from metadata, sync through ordered logs and cursors, treat notifications as hints, make every retry idempotent, and shard by the key every query carries.

Sources used for the designs (primary sources; all designs, numbers and diagrams in this module are original): RFC 5321 (SMTP), RFC 5322 (message format), RFC 9051 (IMAP4rev2), RFC 8620/8621 (JMAP), RFC 7208 (SPF), RFC 6376 (DKIM), RFC 7489 and RFC 9989 (DMARC); RFC 8216 (HLS) and ISO/IEC 23009-1 (MPEG-DASH); Apple's Low-Latency HLS documentation; the Signal Protocol documentation and the WhatsApp security whitepaper; Discord engineering blog posts on storing billions and trillions of messages; Huang et al., "SVE: Distributed Video Processing at Facebook Scale" (SOSP 2017); Netflix Technology Blog on per-title encoding and Netflix Open Connect documentation; Facebook engineering, "Under the hood: Broadcasting live video to millions" (2015); Muthitacharoen, Chen and Mazières, "A Low-Bandwidth Network File System" (SOSP 2001); Tridgell and Mackerras, "The rsync algorithm" (1996); Dropbox engineering blog posts on Magic Pocket and on the Nucleus sync-engine rewrite; the Dropbox and Google Drive public API documentation (change feeds, long-poll/watch); the tus resumable upload protocol and Amazon S3 multipart upload documentation; Disney+ Hotstar's public statements on concurrent-viewer records (2019, 2023); IETF RFC 9725 (WHIP); Kleppmann, *Designing Data-Intensive Applications* (see the linked DDIA notes).